

Piyush Kumar Singh

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EDUCATION

Madhav Institute of Technology and Science

Gwalior, India

B.Tech in Information Technology (AI & Robotics)

Aug. 2023 – Aug. 2027

• CGPA: **8.14/10** — Relevant Coursework: DSA, DBMS, OOP, System Design, OS, Computer Networks

EXPERIENCE

LLM Post-Training Intern

Dec. 2025 – Jan. 2026

Ethara AI (Green Rider Technology LLP)

On-site

- Evaluated LLM responses on curated problem sets sourced from internal platforms, handling **10,000+ daily interactions**.
- Designed structured prompt-response evaluation workflows, identifying incorrect outputs and documenting failure cases with detailed reasoning.
- Performed error analysis and provided corrective feedback, highlighting model reasoning gaps and suggesting improved responses.

Open Source Contributor — Rank #32 / 3414+ (Top 1%)

Aug. 2025 – Nov. 2025

GirlScript Summer of Code (GSSoC)

Remote

- Merged **48+ PRs** across **10+ repos**, contributing **5,000+ LOC**; earned **Top 1%** global ranking.
- Architected **JWT + RBAC** authentication system handling **100+ concurrent users** on a production platform.
- Optimized REST API endpoints and MongoDB query plans, cutting average response time by **40%**.
- Shipped **3 production-ready** backend features end-to-end; conducted peer code reviews improving merge quality.

PROJECTS

CareQueue + Health-Vault | *Node.js, Express.js, MongoDB, Socket.io, JWT, Docker, AES-256*

[GitHub](#)

- Architected **100+ REST APIs** across **13 route modules** and **9 MongoDB models**; compound indexing reduced average DB query latency by **55%**.
- Engineered **AES-256-CBC** encrypted medical records vault with field-level RBAC; secured records for **200+ patient profiles** with zero plaintext exposure.
- Built real-time queue system via **Socket.io WebSocket**, handling **50+ concurrent users** with sub-**200ms** position-update latency under sustained load.
- Designed **JWT + OTP** auth with refresh-token rotation and **rate limiting (100 req/min)**; structured audit logs covering **100%** of write operations for compliance.

Travira — Microservices Safety Platform | *Node.js, Express, Python FastAPI, MongoDB, WebSocket* [GitHub](#) | [Live](#)

- Built **25+ REST APIs** across **5 decoupled microservices**; shared auth middleware reduced per-service auth code by **60%** and cut integration overhead.
- Developed **Python FastAPI** AI scoring service with real-time **WebSocket** streaming; achieved sub-**300ms** threat-score delivery to **30+ concurrent** connected clients.
- Deployed **5 independent services** to **Render** cloud with zero-downtime releases; unified API gateway layer served **3 client platforms** (Android, React, admin) from a single entry point.

TECHNICAL SKILLS

Languages: C++, Java, Python, JavaScript, SQL

Backend: Node.js, Express.js, FastAPI, REST APIs, Microservices, JWT, RBAC, API Rate Limiting

Databases: MongoDB, PostgreSQL, MySQL — Indexing, Query Optimization, Schema Design

Real-Time & Systems: WebSocket, Socket.io, Concurrency Handling, Event-Driven Architecture

Security: AES-256 Encryption, Secure Auth Flows, OTP Verification, Refresh Token Rotation

Tools & DevOps: Docker, Git, GitHub, Postman, Cloud Deployment (Render)

Core CS: Data Structures & Algorithms, OOP, DBMS, System Design, OS, Computer Networks

ACHIEVEMENTS & CERTIFICATIONS

98.55 Percentile among 5L+ participants — Naukri Campus Young Turks 2025.

Hacktoberfest 2025 Supercontributor — high-quality contributions across diverse open-source repos.

Solved **499+ DSA problems** (140E / 292M / 67H) with **800+ GitHub contributions**.

81st Rank (Top 1.5%) — DSA MasterMind Round on Unstop (6,999+ participants).

OCI 2025 Certified Generative AI Professional & OCI AI Foundations Associate — Oracle (Oct. 2025).

Postman API Fundamentals Student Expert (Sep. 2025).